

A Hilbert–Schmidt Criterion for Random Dirichlet Carleson Sequences on the Polydisc

Candidate partial result for arXiv:2402.13645

13 August 2026

Status: candidate partial result, pending human review. For every dimension and every missing parameter $0 < a \leq 1/2$, an explicit double-sum condition makes the random Gramian a Hilbert–Schmidt perturbation of the identity. In particular, the conjectured conclusion holds whenever $N_m \leq C2^{(1/2-\varepsilon)|m|}$. The source’s full finite-mass conjecture is not resolved.

1 Problem and result

Fix $d \geq 1$ and deterministic radii $r_n = (r_n^1, \dots, r_n^d) \in [0, 1]^d$. Let

$$\lambda_n = (r_n^1 e^{i\theta_n^1}, \dots, r_n^d e^{i\theta_n^d}),$$

where the θ_n are independent and uniform on $\mathbb{T}^d$. For $m = (m_1, \dots, m_d) \in \mathbb{N}^d$, put

$$A_m = \{z \in \mathbb{D}^d : 2^{-(m_i+1)} \leq 1 - |z_i| < 2^{-m_i} \text{ for every } i\}, \quad N_m = \#\{n : r_n \in A_m\},$$

and write $|m| = \sum_i m_i$ and $|m \wedge k| = \sum_i \min(m_i, k_i)$.

For $0 < a < 1$, set $\beta = 1 - a$ and let D_d^a be the reproducing-kernel Hilbert space with

$$k_w^{(a)}(z) = \prod_{i=1}^d (1 - \overline{w_i} z_i)^{-\beta}, \quad s_w^{(a)} = k_w^{(a)} / \|k_w^{(a)}\|.$$

The associated discrete measure is

$$\mu_\Lambda = \sum_n \|k_{\lambda_n}^{(a)}\|^{-2} \delta_{\lambda_n}.$$

It is Carleson exactly when $(s_{\lambda_n}^{(a)})_n$ is a Bessel sequence, or equivalently when its Gramian is bounded on ℓ^2 .

Chalmoukis, Dayan, and Lamberti prove that

$$\sum_m N_m 2^{-\beta|m|} < \infty \tag{1}$$

is necessary and sufficient almost surely when $1/2 < a \leq 1$, and conjecture the same equivalence for $0 < a \leq 1/2$ [1].

We conjecture that Corollary 4.5 holds also for $0 < a \leq 1/2$. The random matrix argument used in Theorem 1.1 for the case $a = 0$ can't be used to prove (17) for $0 < a \leq 1/2$, since it requires that the sequence $(N_m 2^{-|m|})_{m \in \mathbb{N}^d}$ decay exponentially. A geometric sufficient condition for Carleson measures for D_d^a in the deterministic setting is only available for $d = 2, 3$: see [17] for the case $0 < a < 1$ and [3] for the case $a = 1$.

Figure 1: The unresolved range stated in arXiv:2402.13645, PDF page 17.

The following theorem extracts the strongest direct Hilbert–Schmidt consequence of the source's second-moment computation.

Theorem 1 (second-moment criterion). *Let Λ be as above.*

1. *If $0 < a < 1/2$ and*

$$\sum_{m, k \in \mathbb{N}^d} N_m N_k 2^{-\beta(|m|+|k|)} 2^{(1-2a)|m \wedge k|} < \infty, \quad (2)$$

then μ_Λ is almost surely a Carleson measure for D_d^a .

2. *If $a = 1/2$ and*

$$\sum_{m, k \in \mathbb{N}^d} N_m N_k 2^{-(|m|+|k|)/2} \prod_{i=1}^d (1 + \min(m_i, k_i)) < \infty, \quad (3)$$

then the same conclusion holds.

In both cases the random Gramian is almost surely a Hilbert–Schmidt perturbation of the identity.

Corollary 2 (square-root-sparse regime). *For every $d \geq 1$ and $0 < a \leq 1/2$, suppose that for some $C < \infty$ and $\varepsilon > 0$,*

$$N_m \leq C 2^{(1/2-\varepsilon)|m|} \quad (m \in \mathbb{N}^d). \quad (4)$$

Then (1) holds and μ_Λ is almost surely Carleson for D_d^a .

Thus the conjecture holds throughout a parameter-independent exponential subregion of its missing range. More generally, (2) and (3) retain anisotropic information through $m \wedge k$ and can apply beyond the pointwise hypothesis (4).

2 Proof

We include the short second-moment argument to make the result independently checkable. In one variable write

$$(1 - \bar{w}z)^{-\beta} = \sum_{\ell \geq 0} c_\ell (\bar{w}z)^\ell, \quad c_\ell \asymp (1 + \ell)^{-a}.$$

If $n \neq j$, independence and uniformity of the angles annihilate all unequal Fourier frequencies. After normalization this gives

$$\mathbb{E} |\langle s_{\lambda_n}^{(a)}, s_{\lambda_j}^{(a)} \rangle|^2 \asymp \prod_{i=1}^d \left[(1 - r_n^i)^\beta (1 - r_j^i)^\beta \sum_{\ell \geq 0} c_\ell^2 (r_n^i r_j^i)^{2\ell} \right]. \quad (5)$$

The expectation factors because the coordinates are independent. The standard power-series estimates are

$$\sum_{\ell \geq 0} (1 + \ell)^{-2a} t^{2\ell} \asymp \begin{cases} (1 - t)^{-(1-2a)}, & 0 < a < 1/2, \\ 1 + \log \frac{1}{1-t}, & a = 1/2. \end{cases} \quad (6)$$

Equations (5)–(6) are precisely Lemma 4.4 of the source.

16

N. CHALMOUKIS, A. DAYAN, AND G. LAMBERTI

Lemma 4.4. *Let $\Lambda = (\lambda_n)_n$ be a random sequence in the polydisc. Then*

$$\mathbb{E}(|S^{(a)}(\lambda_n, \lambda_j)|^2) \simeq \prod_{i=1}^d (1 - r_n^i)^{1-a} (1 - r_j^i)^{1-a} \begin{cases} \frac{1}{(1 - r_n^i r_j^i)^{1-2a}} & 0 \leq a < 1/2 \\ \log \frac{1}{1 - r_n^i r_j^i} & a = 1/2 \\ 1 & 1/2 < a \leq 1 \end{cases}.$$

Figure 2: The source's second-moment estimate, PDF page 16.

If $r_n \in A_m$ and $r_j \in A_k$, then coordinatewise

$$1 - r_n^i \asymp 2^{-m_i}, \quad 1 - r_j^i \asymp 2^{-k_i}, \quad 1 - r_n^i r_j^i \asymp 2^{-\min(m_i, k_i)}. \quad (7)$$

Consequently (5)–(7) imply, for $0 < a < 1/2$,

$$\mathbb{E}|\langle s_{\lambda_n}^{(a)}, s_{\lambda_j}^{(a)} \rangle|^2 \lesssim 2^{-\beta(|m|+|k|)} 2^{(1-2a)|m \wedge k|}, \quad (8)$$

and at the endpoint,

$$\mathbb{E}|\langle s_{\lambda_n}^{(1/2)}, s_{\lambda_j}^{(1/2)} \rangle|^2 \lesssim 2^{-(|m|+|k|)/2} \prod_{i=1}^d (1 + \min(m_i, k_i)). \quad (9)$$

Let $G = (\langle s_{\lambda_n}^{(a)}, s_{\lambda_j}^{(a)} \rangle)_{n,j}$ and $H = G - I$. Tonelli's theorem, grouping ordered pairs by dyadic cells, gives from (8) or (9)

$$\mathbb{E}\|H\|_{\text{HS}}^2 = \sum_{n \neq j} \mathbb{E}|\langle s_{\lambda_n}^{(a)}, s_{\lambda_j}^{(a)} \rangle|^2 < \infty. \quad (10)$$

Hence the nonnegative random variable $\|H\|_{\text{HS}}^2$ is finite almost surely. Therefore $G = I + H$ is bounded on ℓ^2 almost surely. For every $f \in D_d^a$,

$$\int |f|^2 d\mu_\Lambda = \sum_n |\langle f, s_{\lambda_n}^{(a)} \rangle|^2 \leq \|G\| \|f\|^2,$$

first on finite subsystems and then by monotone convergence. This proves Theorem 1.

For Corollary 2, first suppose $0 < a < 1/2$ and put $\alpha = 1 - 2a$. Since

$$|m \wedge k| \leq \frac{|m| + |k|}{2}, \quad \beta - \frac{\alpha}{2} = \frac{1}{2},$$

each summand in (2), under (4), is at most $C^2 2^{-\varepsilon(|m|+|k|)}$. Its sum over $\mathbb{N}^d \times \mathbb{N}^d$ converges. At $a = 1/2$, a summand in (3) is at most

$$C^2 2^{-\varepsilon(|m|+|k|)} (1 + |m| + |k|)^d,$$

which is again summable. Finally,

$$N_m 2^{-(1-a)|m|} \leq C 2^{-(1/2-a+\varepsilon)|m|},$$

so (1) also holds for all $0 < a \leq 1/2$.

3 Why this does not settle the full conjecture

Finite mass alone permits N_m almost as large as $2^{(1-a)|m|}$ on a sparse set of cells, far above the square-root scale. For $a \leq 1/2$, the factor $2^{(1-2a)|m \wedge k|}$ (or its endpoint logarithm) can then make (10) diverge. Divergence of the Hilbert–Schmidt expectation is only failure of this proof, not evidence that the Gramian is unbounded.

Eight focused upgrade routes were checked. Schur-product reduction cannot assign arbitrary product mass to finitely many coordinatewise summable classes. Finite mass does imply an almost-sure finite separated coloring, but no available product-Dirichlet theorem turns that fact plus finite mass into the desired embedding. Iterating the one-variable random theorem would require an operator-valued version with non-scalar weights. The known bi-tree and tri-tree criteria reduce the question to uniform quadratic energy estimates over all rectangles, which finite total mass does not by itself supply; the cited theory also has a dimension-four obstruction [2]. Direct same-shell simulations revealed no growing Gram norm and therefore no counterexample. Finally, a beta-mixture representation of the kernel does not yield Hardy estimates uniform enough to integrate.

These obstructions leave the finite-mass conjecture credible but unresolved. Theorem 1 is a rigorous intermediate result because it isolates the exact overlap energy that the direct second-moment method can control.

Since all the kernels involved are invariant under rotations, Λ generates for all ω in Ω a finite measure for D_d^a if and only if

$$\sum_{n \in \mathbb{N}} \|k_{r_n}^{(a)}\|^{-2} = \sum_{m \in \mathbb{N}^d} N_m 2^{-(1-a)|m|} < \infty. \quad (16)$$

Recall that for $d = 1$ and $0 < a \leq 1$, (16) is also sufficient for Λ to generate a Carleson measure for D^a , see [10] Theorem 1.4]. Thanks to Lemma 4.4, this can be seen to be true also in the multi-variable case, though our argument covers only the case $1/2 < a \leq 1$. Indeed, if (16) holds, thanks to Lemma 4.4, $1/2 < a \leq 1$, one obtains that

$$\begin{aligned} \mathbb{E} \left(\sum_{n \neq j} |S^{(a)}(\lambda_n, \lambda_j)|^2 \right) &= \sum_{n \neq j} \mathbb{E} (|S^{(a)}(\lambda_n, \lambda_j)|^2) \\ &\simeq \sum_{n \neq j} \|k_{r_n}^{(a)}\|^{-2} \|k_{r_j}^{(a)}\|^{-2} < \infty, \end{aligned}$$

hence the Gram matrix of Λ in D_d^a

$$G_\Lambda^a := (S^{(a)}(\lambda_n, \lambda_j))_{n,j}$$

is almost surely a Hilbert–Schmidt perturbation of the identity. In particular,

Figure 3: The source’s Hilbert–Schmidt proof above the endpoint, PDF page 16. Theorem 1 supplies the corresponding exact second-moment conditions below and at the endpoint.

References

- [1] N. Chalmoukis, A. Dayan, and G. Lamberti, *Random Carleson Sequences for the Hardy Space on the Polydisc and the Unit Ball*, arXiv:2402.13645; J. Funct. Anal. 287 (2024), 110477.
- [2] P. Mozolyako, G. Psaromiligkos, A. Volberg, and P. Zorin-Kranich, *Carleson embedding on the tri-tree and on the tri-disc*, Rev. Mat. Iberoam. 38 (2022), 2069–2116; arXiv:2001.02373.